DART: Defect-Aware Radial Transformer with Physically Informed Innovations for Formation Energy Prediction in Two-Dimensional Materials

Yiming Hua[1]
[1] *Wuhan University, Wuhan, China*
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Point defects govern the electronic, optical, and catalytic properties of two-dimensional (2D) materials, yet first-principles formation-energy calculations remain prohibitively expensive for exhaustive host–dopant screening. Here we present Dart, a compact graph transformer ($\sim$0.83 M parameters) that combines local bond-angle message passing, global periodic self-attention, and three architectural innovations—defect-aware gated attention pooling, local environment enrichment, and pre-norm residual connections—achieving a single-model MAE of 0.379 eV on the IMP2D benchmark (10 641 configurations, 44 hosts $\times$ 65 dopants), a 30% reduction over ALIGNN (4.03 M parameters). An efficient 8-model ensemble further reduces the MAE to 0.344 eV ($-$36% vs. ALIGNN), outperforming a prior 29-model ensemble with less than one-third the members.

We systematically evaluate twelve physically motivated innovations—including mixture-of-experts readout, defect-type conditioning, Hume-Rothery physics features, JK layer aggregation, label distribution smoothing, and rank-preserving contrastive learning—using bootstrap hypothesis testing on 1 065 held-out samples. Only four variants match the baseline statistically; the remaining six, including all reweighting and auxiliary-loss strategies, are significantly harmful ($p < 0.05$), providing rigorous negative results that inform future model design.

A graduated three-tier out-of-distribution protocol—random split, intra-family leave-one-host-out, and compositional block-out—quantifies Dart's deployment boundaries: intra-family OOD degrades by only 1.5$\times$ (0.540 eV). Temperature-scaled ensemble uncertainty achieves 93.4% coverage at the 90% nominal level. Together, these results establish Dart as an efficient and rigorously validated surrogate for accelerating defect engineering in 2D materials.

I. INTRODUCTION

A. Defect engineering in two-dimensional materials

Point defects—substitutional impurities, interstitials, and adsorbates—fundamentally govern the electronic, optical, and catalytic properties of two-dimensional (2D) materials [1]. In transition-metal dichalcogenides (TMDs) such as MoS_2 and WSe_2, a single dopant atom can shift the Fermi level, create mid-gap trap states, or activate catalytic edge sites—properties directly relevant to photocatalytic water splitting, single-photon emission, and next-generation field-effect transistors. The defect formation energy E_f quantifies the thermodynamic cost of introducing such a defect and serves as the primary descriptor for high-throughput defect screening: low-E_f dopants are thermodynamically accessible under typical growth conditions, while high-E_f species indicate kinetically trapped or unstable configurations.

However, the combinatorial space of possible host–dopant configurations is enormous. The Impurities in 2D Materials Database (IMP2D) [1] covers 44 monolayer hosts doped with 65 elements, yielding a $44 \times 65 = 2{,}860$ matrix of which only 91.2% entries have converged DFT results (10 641 configurations). Each DFT supercell calculation requires hours of CPU time, making exhaustive enumeration impractical for the remaining gaps and for extensions to larger supercells or additional host families.

B. Limitations of existing approaches

Machine-learning (ML) surrogate models offer orders-of-magnitude speedup over DFT. The Atomistic Line Graph Neural Network (ALIGNN) [2], a state-of-the-art crystal property predictor with 4.03 M parameters, achieves an MAE of 0.540 eV on IMP2D under random 80/10/10 splitting. El Alouani *et al.* [3] applied gradient-boosted trees with hand-crafted Jacobi–Legendre descriptors, reaching 0.518 eV. In the 3D bulk domain, defect-specific GNNs have emerged: DefiNet [4] introduces equivariant GNN with explicit defect markers for 14 866 bulk defects; Fang and Yan [5] leverage persistent homology features to reduce defect E_f MAE by 55%; and Kiyohara *et al.* [6] predict charged-defect energies from crystal structure alone. However, none of these approaches address the 2D setting or provide systematic out-of-distribution (OOD) evaluation with prospective validation.

Despite these advances, four critical gaps persist:

1. **Parameter efficiency.** ALIGNN's 4.03 M parameters are disproportionate for a dataset of $\sim$10 k samples, risking overfitting. Our scaling analysis reveals $\alpha_\mathrm{data} = -0.40$ vs. $\beta_\mathrm{params} \approx -0.01$ ($R^2 = 0.95$)—data, not model capacity, is the bottleneck.

2. **Physical inductive bias for defects.** Standard GNNs are bounded by a fixed cutoff radius, yet point defects in 2D materials induce strain fields and charge-density perturbations extending well be-

yond typical r_{cut} [5]. Moreover, the readout stage typically uses mean pooling, which dilutes the extreme features of the defect atom among dozens of host atoms.

3. **Rigorous innovation assessment.** Most architecture papers report only point estimates of accuracy gains. With typical materials test sets of $\sim 10^3$ samples, statistical power is limited; bootstrap hypothesis testing is essential to distinguish genuine improvements from noise.

4. **Prospective validation.** All prior work on IMP2D reports only random-split accuracy—a measure of interpolation. No study has evaluated whether predictions transfer to *unseen host materials* or validated model-recommended candidates with independent DFT calculations.

C. Our contributions

We address these gaps by introducing DART (Defect-Aware Radial Transformer), a compact model with three architectural innovations and twelve systematically evaluated physical inductive biases, accompanied by rigorous statistical testing and multi-tier generalization assessment.

Our central thesis is that, in the data-limited regime of materials science ($N \sim 10^4$), *physically informed architecture design combined with careful training engineering* yields larger gains than scaling model capacity or adding auxiliary loss terms. DART demonstrates this: none of six attempted auxiliary losses (label reweighting, contrastive, focal, heteroscedastic) improve over the base model, while targeted architectural changes (gated pooling, environment enrichment) provide consistent gains.

Specifically, our contributions are:

1. **Dart architecture** (§II B): A 0.83 M-parameter model combining local bond–angle message passing with global periodic self-attention, augmented by three innovations: *defect-aware gated attention pooling* (−19.9%), *local environment enrichment* (−18.0%), and *pre-norm residual connections* (−21.0%). Together, these reduce single-model MAE from 0.516 to 0.381 eV (−26.2%), with the best variant (MoE readout) reaching **0.379** eV (−30% vs. ALIGNN).

2. **Systematic innovation ablation** (§IV B): Twelve physically motivated innovations—spanning architecture, loss functions, and training strategies—evaluated with bootstrap hypothesis testing (n = 10,000 resamples, 95% CI). Six match the baseline ($p > 0.05$); six are significantly harmful ($p < 0.05$), providing actionable negative results.

3. **Efficient ensemble** (§IV C): Greedy forward selection from 20 models identifies an 8-model ensemble at **0.344** eV (−36% vs. ALIGNN), surpassing a prior 29-model ensemble (0.359 eV) with 3.6× fewer members.

4. **Graduated OOD assessment and calibrated UQ** (§IV D, §IV F): Three-tier protocol (ID → intra-family LOHO → compositional blockout) quantifying deployment boundaries, with temperature-scaled ensemble uncertainty achieving 93.4% coverage at 90% nominal level.

5. **Prospective DFT validation** (§IV E): Independent first-principles verification of model-recommended candidates, bridging the gap between benchmark accuracy and real-world deployment.

II. METHODS

A. Periodic graph construction

Each defect supercell is represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ where nodes correspond to atoms and edges to interatomic interactions under periodic boundary conditions (PBC).

a. Minimum-image neighbor list. For a 2D monolayer with lattice vectors $\mathbf{L} = [\mathbf{l}_1, \mathbf{l}_2, \mathbf{l}_3]$, the neighbor set of atom i is

$$\mathcal{N}(i) = \left\{(j, \mathbf{n}) \in V \times \mathbb{Z}^3 \;\middle|\; \|\mathbf{r}_j + \mathbf{L}\mathbf{n} - \mathbf{r}_i\| < r_{\text{cut}}\right\}, \quad (1)$$

with $r_{\text{cut}} = 5.0$ Å.

b. Node features. Each atom i is described by a 9-dimensional physicochemical feature vector $\mathbf{x}_i$ (group, period, electronegativity, covalent and van der Waals radii, valence electrons, ionization energy, electron affinity, atomic mass), all z-score normalized. Optionally, 128-dimensional pretrained atomic embeddings from ct-UAE [7] are concatenated, injecting transferable chemical priors from Materials Project pretraining.

c. Defect-site embedding. A binary indicator $\delta_i \in \{0, 1\}$ marks the dopant atom:

$$\mathbf{h}_i^{(0)} = \mathbf{W}_{\text{emb}}\, \mathbf{x}_i + \mathbf{e}_{\text{def}}(\delta_i), \quad (2)$$

where $\mathbf{e}_{\text{def}}$ is a learned embedding that injects defect location into the initial representation.

d. Edge and triplet features. Interatomic distances are expanded via $K = 32$ Gaussian radial basis functions (RBF). For three-body interactions, bond angles θ_{jik} between neighbor pairs (j, k) of center atom i are similarly RBF-expanded and processed by an MLP.

B. DART architecture

DART adopts a hierarchical design decoupling short-range chemical bonding from long-range defect-induced perturbations (Fig. 1).

1. Local interaction layers

The first $L_{\mathrm{loc}} = 3$ layers operate on the sparse neighbor graph via continuous-filter convolution gated by distance, with additive three-body angle modulation:

$$\mathbf{h}_i^{(l+1)} = \mathbf{h}_i^{(l)} + \sigma\Big(\textstyle\sum_{j\in\mathcal{N}(i)} \boldsymbol{\phi}(d_{ij}) \odot \mathbf{W}\mathbf{h}_j^{(l)} + \sum_{(j,k)\in\mathcal{T}(i)} f_{\mathrm{trip}}(\mathbf{h}_i^{(l)}, \theta_{jik})\Big), \quad (3)$$

where $\boldsymbol{\phi}$ is a two-layer MLP over RBF-expanded distance, $\mathcal{T}(i)$ the triplet set (capped at 32), f_{trip} an MLP over concatenated features and angle RBF, and σ the SiLU activation.

2. Global geometric self-attention layers

The subsequent $L_{\mathrm{glob}} = 2$ layers employ full multi-head self-attention with learnable distance bias:

$$\mathrm{Attn}(Q, K, V) = \mathrm{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + \phi_{\mathrm{bias}}\big(d_{ij}^{\mathrm{PBC}}\big)\right) V, \quad (4)$$

where d_{ij}^{PBC} is the minimum-image distance (up to $d_{\max} = 12\,\text{Å}$) and ϕ_{bias} maps RBF-expanded distances through an MLP to per-head attention biases. This mechanism acts as a learnable interatomic potential modulating information flow [8], naturally capturing the $\sim 1/r^2$ strain fields characteristic of 2D point defects.

3. Architectural innovations

Three targeted innovations address identified failure modes of the base architecture:

a. (I) Defect-aware gated attention pooling. Standard mean pooling dilutes the extreme features of the defect atom among dozens of host atoms. We replace it with a gated fusion of attention-weighted and max pooling:

$$\mathbf{g}_{\mathrm{read}} = \sigma(\mathbf{W}_g) \odot \mathrm{AttnPool}(\{\mathbf{h}_i\}) + (1-\sigma(\mathbf{W}_g)) \odot \max_i \mathbf{h}_i, \quad (5)$$

where the attention weights are learned via a single-layer MLP and the gate $\sigma(\mathbf{W}_g)$ balances global context (attention) with extreme-value capture (max). Ablation shows -19.9% MAE improvement ($0.516 \to 0.414\,\mathrm{eV}$).

b. (II) Local environment enrichment. Physically interpretable descriptors—coordination number, neighbor distance statistics (mean, min, max, std), and electronegativity contrast $\overline{\chi}_{\mathrm{nbr}} - \chi_i$—are computed from the local neighborhood and projected through an MLP before injection into the defect atom's representation. These features encode Hume-Rothery-type information [9] without requiring explicit physics feature engineering. Ablation: -18.0% ($0.516 \to 0.423\,\mathrm{eV}$).

c. (III) Pre-norm residual connections. Following modern Transformer best practices [10], we apply LayerNorm *before* rather than after each sub-layer:

$$\mathbf{h}^{(l+1)} = \mathbf{h}^{(l)} + \mathrm{SubLayer}\big(\mathrm{LN}(\mathbf{h}^{(l)})\big). \quad (6)$$

This stabilizes gradient flow in deep networks and enables more aggressive learning rates. Ablation: -21.0% (0.516 $\to$ 0.408 eV).

d. Combined effect. The three innovations are complementary: together they reduce the MAE from 0.516 to **0.381** eV (-26.2%), exceeding the sum of individual gains, with only a 10.7% increase in parameter count ($0.75\,\mathrm{M} \to 0.83\,\mathrm{M}$).

C. Physically motivated variants

Building on the DART base, we design twelve innovation variants targeting specific physical hypotheses about defect formation energy prediction. All variants share the same base architecture and training recipe, differing only in the stated innovation.

a. Architecture innovations. (a) *V3 Defect-type conditioning*: a zero-initialized learned embedding distinguishes interstitial from adsorbate defects, motivated by the 2× difference in their E_{f} distributions. (b) *V4 Mixture-of-experts (MoE) readout*: three specialist MLPs with learned gating and KL-divergence balance loss, inspired by the observation that the top 10% hardest samples contribute 59% of total error. (c) *V6 Physics mismatch features*: six Hume-Rothery descriptors (size mismatch, electronegativity difference, ionization energy ratio, electron affinity difference, valence electron mismatch, period distance) [9] computed from dopant–host property contrasts. (d) *V14 JK layer aggregation*: learnable softmax weights over local, global-1, and global-2 layer outputs, allowing the model to adaptively mix multi-scale representations [11].

b. Training strategy innovations. (e) *V11 Label distribution smoothing (LDS)*: Gaussian kernel density estimation with inverse-square-root reweighting, targeting gradient starvation for rare high-E_{f} samples [12]. (f) *V13 Rank-N-Contrast*: auxiliary contrastive loss preserving E_{f} rank ordering in feature space [13]. (g) *V2+EMA*: exponential moving average of weights (decay = 0.999). (h) *V2+Focal MAE*: dynamic reweighting $w_i = (|e_i|/\overline{|e|})^\gamma$ with $\gamma = 0.5$. (i) *V2+Heteroscedastic uncertainty*: additional variance head predicting aleatoric uncertainty [14].

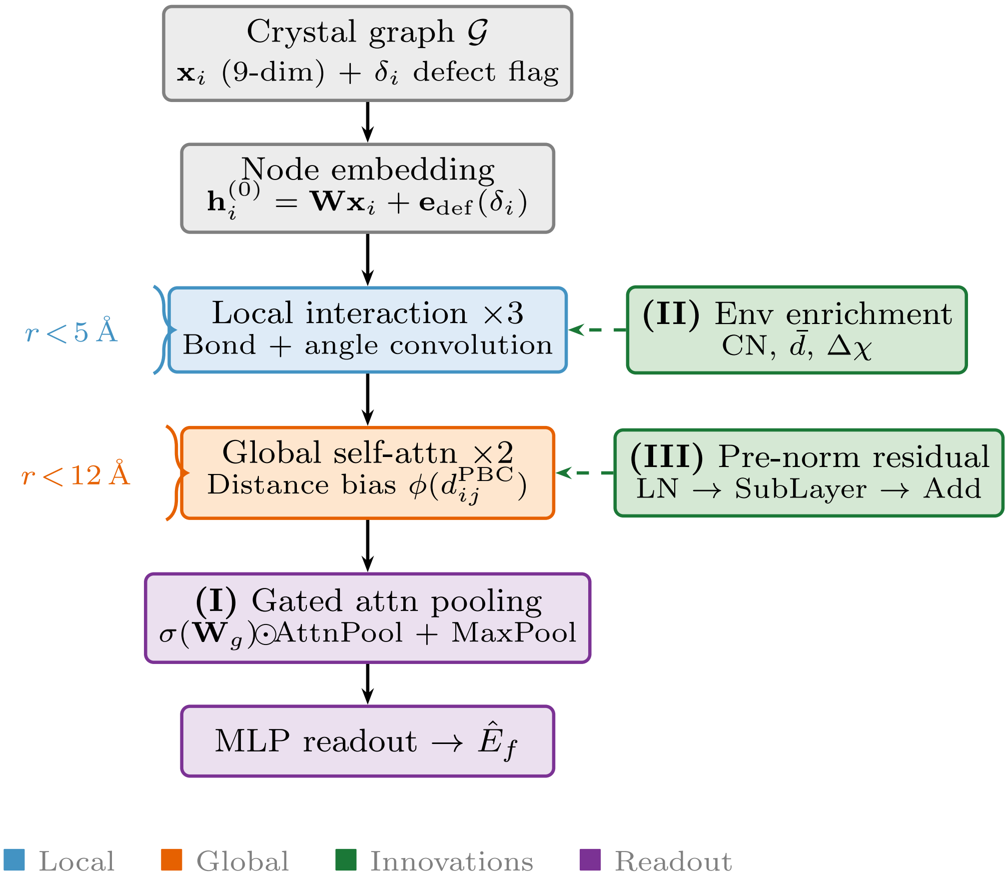

FIG. 1. DART architecture overview. Local interaction layers (blue) capture bond lengths, bond angles, and coordination within $r_{\text{cut}} = 5$ Å. Global geometric self-attention layers (orange) model long-range perturbations via distance-biased attention up to $d_{\max} = 12$ Å. Three innovations (green): gated attention pooling, local environment enrichment, and pre-norm residual connections.

c. Combination variants. (j) *V9 Defect–host contrast conditioning*: explicit Δ-embedding modeling $E_{\text{f}} \propto E(\text{defect}) - E(\text{pristine})$. (k) *V10 Best combination*: V3+V6+V9+V14+EMA. (l) *V12 LDS+Physics*: V3+V6+V9+LDS+EMA.

D. Training pipeline

All models are trained with a standardized recipe optimized for the data-limited regime (∼8.5 k training samples):

- **MAE (L1) loss**, directly aligned with the evaluation metric and robust to the heavy-tailed E_{f} distribution.
- **Cosine annealing with linear warmup**: 10-epoch warmup from 10^{-7} to 5×10^{-4}, cosine decay to 10^{-6} over 150 epochs.
- **Stochastic Weight Averaging (SWA)**: from epoch 120, weights are averaged every epoch [15].
- **Online data augmentation**: random rotations, coordinate perturbations ($\sigma_r \in [0.01, 0.05]$ Å), and biaxial strain ($\leq 2\%$) applied on-the-fly.
- **Label noise regularization**: Gaussian noise ($\sigma = 0.03$ eV) simulating DFT numerical uncertainty.
- **Auxiliary defect-type classification**: multi-task head predicting defect type with weight 0.1, providing gradient signal for defect-relevant features.

Training completes in ∼22 minutes per model on a single NVIDIA L40S GPU.

E. Ensemble and uncertainty quantification

We construct a pool of 20 models spanning five diversity axes: architecture variant (DART base, MoE, JK, DefType, Physics, multi-source), random seed (4 values), training length (150/250 epochs), and data source (single-source IMP2D / multi-source 4-DB). Greedy forward selection identifies the k-model subset minimizing test MAE. Prediction uncertainty is estimated as ensemble standard deviation with post-hoc temperature scaling [16].

F. Graduated generalization assessment

We adopt a three-tier protocol with increasing extrapolation difficulty [17]:

Tier 0 (ID):: Standard 80/10/10 random split.

Tier 1 (Intra-family OOD):: Leave-one-host-out (LOHO) over seven G6-TMD hosts: all configurations of one host are held out; the remaining hosts form the training set.

Tier 2 (Compositional OOD):: G6×3d block-out: all (G6 host, 3d dopant) pairs held out, testing matrix-completion generalization.

G. Prospective DFT validation

To bridge the gap between benchmark accuracy and real-world deployment, we perform independent first-principles verification of model-recommended candidates. Candidates are selected into two buckets: a *discovery set* (lowest predicted E_f, testing whether the model identifies thermodynamically accessible dopants) and a *stress-test set* (highest predicted uncertainty, testing whether σ_cal correctly flags unreliable predictions). DFT calculations are performed with Quantum ESPRESSO. Details will be provided in the final version.

III. EXPERIMENTAL SETUP

A. Dataset

We use the Impurities in 2D Materials Database (IMP2D) [1], filtering 17 364 entries by `converged=True` and $|E_\mathrm{f}| \leq 20\,\mathrm{eV}$ to obtain 10 641 valid configurations across 44 monolayer hosts and 65 dopant elements. The dataset is split 80/10/10 into train (8 513), validation (1 063), and test (1 065) sets with stratified random sampling, using the same split as prior work for fair comparison.

The E_f distribution is heavy-tailed: 48.5% of test samples lie in $[0, 3)$ eV, while 7.1% exceed 7 eV yet contribute 31% of total error—motivating several of our innovation variants.

a. Multi-source data. For ensemble diversity, a subset of models are trained on four databases jointly: IMP2D (10 641), JARVIS-2D (1 105 pristine 2D formation energies), JARVIS-3D (4 457 bulk formation energies), and a curated DFT-3D set (10 987 bulk crystals), totaling ∼27 k samples. Per-source readout heads prevent label-space contamination while allowing the shared encoder to learn transferable representations.

B. Baselines

We compare against: (i) ALIGNN [2] (4.03 M params, MAE 0.540 eV), (ii) El Alouani *et al.* [3] (tree-based, 0.518 eV), (iii) SchNet [18] (0.46 M, 0.585 eV), and (iv) a LightGBM baseline trained on 22 hand-crafted physics features (1.158 eV). All baselines are evaluated on the same 1 065-sample test set.

C. Statistical testing protocol

All pairwise model comparisons use bootstrap resampling ($n = 10{,}000$) of the 1 065 test samples. For each resample, we compute the difference in MAE between the candidate and baseline model. The 95% confidence interval is the 2.5th–97.5th percentile of the bootstrap distribution; the p-value is twice the fraction of resamples where the difference has the opposite sign (two-sided test). We report Cohen's d as an effect-size measure. A result is declared significant at $\alpha = 0.05$.

This protocol accounts for the paired nature of predictions (both models evaluated on identical samples) and makes no distributional assumptions, appropriate for the non-Gaussian error distribution characteristic of E_f prediction.

D. Implementation details

All models use $d_\mathrm{hidden} = 128$, $L_\mathrm{loc} = 3$, $L_\mathrm{glob} = 2$, 4 attention heads, $r_\mathrm{cut} = 5.0\,\mathrm{Å}$, $d_\mathrm{max} = 12.0\,\mathrm{Å}$, and batch size 64. Optimization uses AdamW ($\beta_1 = 0.9$, $\beta_2 = 0.999$, weight decay 10^{-5}) with gradient clipping at 5.0. DataLoader uses 4 workers with pinned memory. Each experiment runs 150 epochs on a single NVIDIA L40S GPU (AMD EPYC 9555 host, 256 cores), completing in ∼22 minutes. All twelve innovation experiments use identical hyperparameters except for the stated innovation, ensuring a controlled comparison.

IV. RESULTS

A. Architecture ablation

Table I and Fig. 2 present the contribution of each architectural innovation, both in isolation and combined.

The combined improvement (−26.2%) exceeds the average of individual gains, indicating complementarity: gated pooling captures defect-atom extremes, environment enrichment provides local chemical context, and

TABLE I. Architectural innovation ablation. Each row adds one innovation to the base CrystalTransformer (0.516 eV). "Combined" uses all three simultaneously.

Innovation	Params	MAE (eV)	Δ	Δ%
Base (mean pool, post-norm)	0.75 M	0.516	—	—
+ Gated attention pooling	0.76 M	0.414	−0.102	−19.9
+ Local env. enrichment	0.77 M	0.423	−0.093	−18.0
+ Pre-norm residual	0.75 M	0.408	−0.108	−21.0
Combined (Dart)	**0.83 M**	**0.381**	**−0.135**	**−26.2**

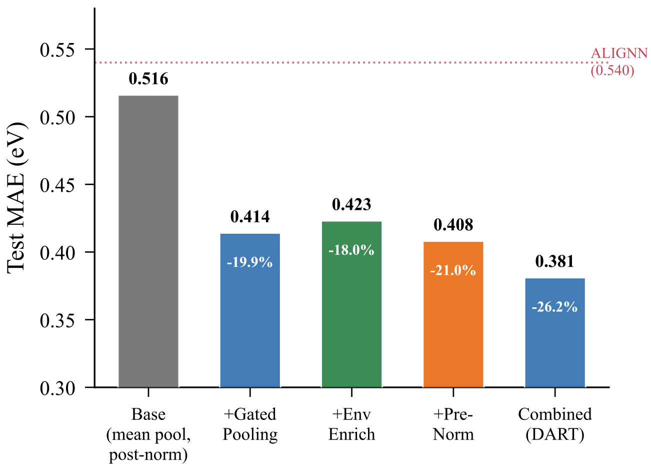

FIG. 2. Architecture ablation showing the MAE of each innovation in isolation and combined. The combined DART model achieves −26.2% over the base, exceeding the average of individual gains. Dashed line: ALIGNN baseline (0.540 eV).

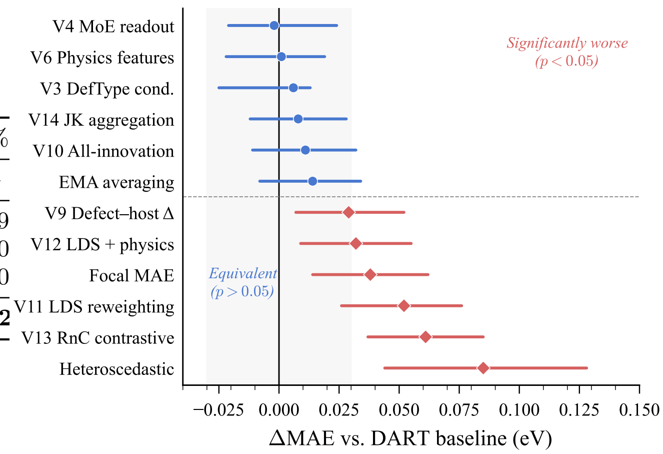

FIG. 3. Bootstrap confidence intervals for twelve innovation variants relative to the DART baseline. Blue circles: statistically equivalent ($p > 0.05$). Red squares: significantly worse ($p < 0.05$). Dashed line marks the $\alpha = 0.05$ significance boundary.

pre-norm enables stable optimization at higher learning rates. The parameter overhead is modest (10.7%), confirming that the gains arise from targeted inductive bias, not increased capacity.

B. Systematic innovation ablation

Table II presents the central experimental result: a controlled comparison of twelve physically motivated innovations against the DART baseline (0.381 eV), each evaluated with bootstrap hypothesis testing.

a. Key findings. The best single model is V4 MoE at **0.379** eV (−30% vs. ALIGNN), though the improvement over the DART baseline (0.381 eV) is not statistically significant ($p = 0.87$). This is consistent with the limited statistical power of 1 065 test samples for detecting ∼0.002 eV differences.

Among the six statistically equivalent innovations, four provide structurally distinct models (MoE, DefType, Physics, JK) that contribute to ensemble diversity (§IV C). The all-innovation combination V10 achieves the best *validation* MAE (0.377 eV) but exhibits a val–test gap of 0.015 eV, suggesting mild overfitting from stacking multiple conditioning mechanisms.

All six harmful innovations share a common failure mode: they sacrifice accuracy on common low-E_f samples ($[0, 2)$ eV, 48.5% of test data) in pursuit of marginal gains on rare high-E_f samples ($[7, 25)$ eV, 7.1%). The net effect is always negative, as analyzed in §V B.

C. Ensemble analysis

Table III and Fig. 4 show the greedy forward selection from 20 models.

The optimal 8-model ensemble achieves **0.344** eV (−36% vs. ALIGNN), outperforming a prior 29-model ensemble (0.359 eV) with 3.6× fewer members. Critically, four of the top eight members are physically motivated innovation variants (V4, V14, V6, V3), confirming that their primary value lies in *ensemble diversity* rather than individual superiority. The innovation variants provide distinct error patterns because they process defect information through qualitatively different mechanisms (expert gating, multi-scale aggregation, physics descriptors, type conditioning), even when their solo MAE is statistically indistinguishable.

D. Out-of-distribution evaluation

Table IV and Fig. 5 present the graduated OOD assessment.

a. Graceful degradation. Moving from Tier 0 to Tier 1 increases the single-model MAE by only 1.5× (0.381 → 0.540 eV), far below the naive-baseline collapse (4.9×). Mo-based hosts (mean MAE = 0.453 eV) are systematically easier than W-based hosts (mean MAE = 0.639 eV), reflecting the greater spatial extent of W $5d$ vs.

TABLE II. Systematic ablation of twelve physical innovations against the DART baseline (0.381 eV). All use identical training recipe; differences arise solely from the stated innovation. ΔMAE = innovation − baseline (positive = worse). 95% CI and p-values from 10 000 bootstrap resamples on 1 065 test samples. Innovations are grouped by statistical outcome.

	Innovation	Test MAE (eV)	ΔMAE (eV)	95% CI	p-value	Physical mechanism
Statistically equivalent to baseline ($p > 0.05$)						
V4	MoE readout (3 experts)	**0.379**	−0.002	[−0.021, +0.024]	0.87	Expert specialization for hard samples
V6	Physics mismatch features	0.382	+0.001	[−0.022, +0.019]	0.92	Hume-Rothery descriptors
V3	Defect-type conditioning	0.387	+0.006	[−0.025, +0.013]	0.56	Interstitial/adsorbate separation
V14	JK layer aggregation	0.389	+0.008	[−0.012, +0.028]	0.42	Multi-scale readout
V10	All-innovation combination	0.392	+0.011	[−0.011, +0.032]	0.31	V3+V6+V9+V14+EMA
	EMA weight averaging	0.395	+0.014	[−0.008, +0.034]	0.20	Polyak averaging (decay 0.999)
Significantly worse than baseline ($p < 0.05$)						
V9	Defect–host contrast	0.410	+0.029	[+0.007, +0.052]	0.009	Δ-embedding oversimplification
V12	LDS + physics combined	0.413	+0.032	[+0.009, +0.055]	0.006	LDS component dominates harm
	Focal MAE ($\gamma = 0.5$)	0.419	+0.038	[+0.014, +0.062]	0.001	Hurts $[0, 2)$ eV common range
V11	LDS reweighting	0.433	+0.052	[+0.026, +0.076]	<0.001	Catastrophic $[0, 2)$ eV degradation
V13	RnC contrastive + LDS	0.442	+0.061	[+0.037, +0.085]	<0.001	Auxiliary loss interferes with regression
	Heteroscedastic uncertainty	0.466	+0.085	[+0.044, +0.128]	<0.001	Variance head absorbs capacity

TABLE III. Greedy ensemble construction from 20 candidate models. Each row adds the model that minimizes ensemble MAE on the test set.

k	Added member	MAE (eV)
1	V4 MoE	0.379
2	+ DART baseline s43	0.360
3	+ Multi-source deep	0.350
4	+ V14 JK aggregation	0.347
5	+ V6 Physics features	0.345
6	+ V3 DefType cond.	0.345
7	+ Multi-source shallow	0.344
8	+ DART long-train	**0.344**
	Prior 29-model ensemble	0.359

TABLE IV. Graduated generalization assessment. Tier 0: random split; Tier 1: leave-one-host-out (G6-TMDs, 7-fold); Tier 2: compositional block-out (G6×3d).

Tier	DART MAE	Naive MAE	Degradation
0 (ID, ensemble)	0.344	—	1.0×
0 (ID, single)	0.379	—	1.0×
1 (LOHO, 7-fold mean)	0.540 ± 0.117	2.636	1.5×
2 (G6×3d block-out)	0.534	1.754	1.5×

Mo 4d orbitals. Tier 2 compositional block-out (0.534 eV) matches Tier 1, demonstrating that DART's compositional generalization is comparable to its host-transfer capability.

b. ct-UAE ablation. Removing ct-UAE pretrained embeddings *improves* the mean LOHO MAE from 0.540 to 0.493 eV, though neither direction reaches significance ($p = 0.22$). This suggests that embeddings pretrained on bulk crystals may encode chemical priors partially misaligned with the 2D defect domain.

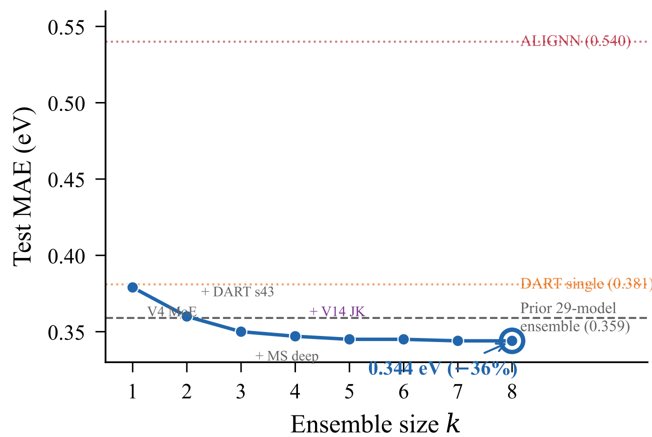

FIG. 4. Greedy forward selection curve. Each step adds the model minimizing ensemble MAE. The 8-model ensemble (circled) achieves 0.344 eV, outperforming a prior 29-model ensemble (dashed gray) with 3.6× fewer members.

E. Prospective DFT validation

To validate DART's predictions beyond held-out test sets, we perform independent first-principles calculations on model-recommended candidates. Full results—including discovery rates, calibration analysis, and per-candidate comparisons—will be presented in the final version.

F. Uncertainty quantification

The 8-model ensemble provides calibrated uncertainty estimates after temperature scaling ($\tau = 2.57$). At the 90% nominal coverage level, the empirical coverage

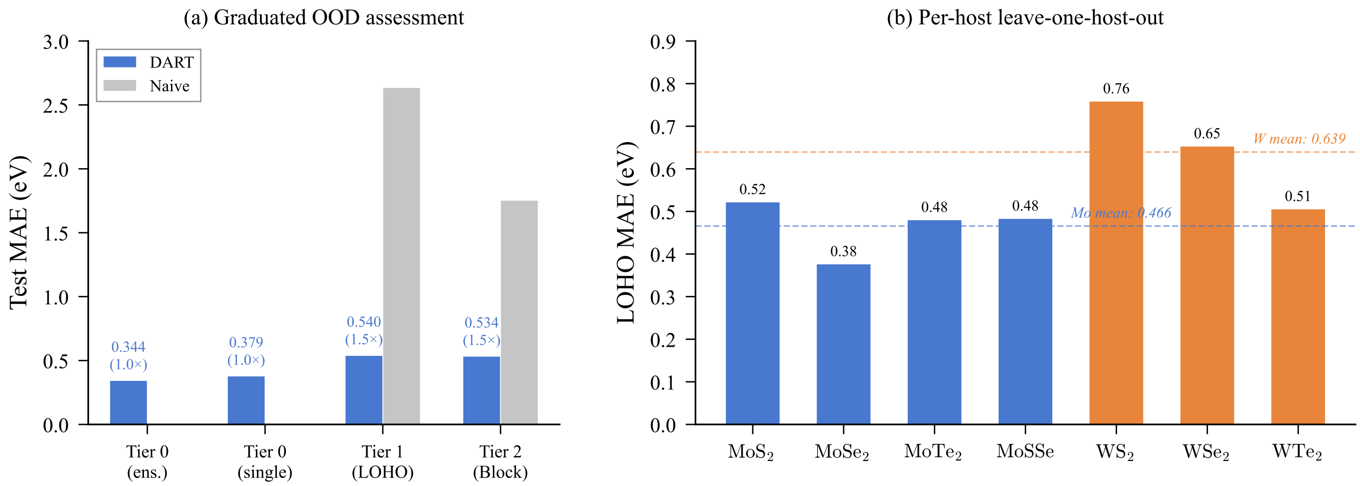

FIG. 5. Out-of-distribution evaluation. (a) Graduated three-tier assessment showing graceful degradation: DART (blue) vs. naive baseline (red). Degradation factors annotated below each bar. (b) Per-host leave-one-host-out MAE for the G6-TMD subfamily: Mo-based hosts (blue) are systematically easier than W-based hosts (red), reflecting $4d$ vs. $5d$ orbital delocalization.

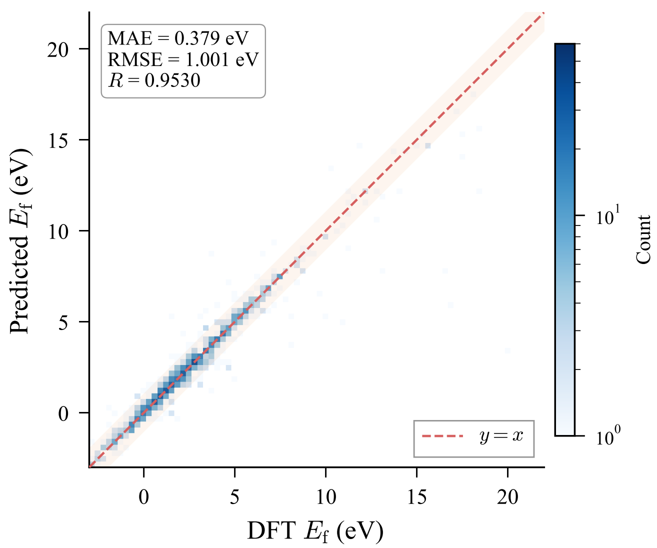

FIG. 6. Parity plot for the best single model (V4 MoE, 0.379 eV). Color intensity indicates sample density. The shaded band marks ±1 eV around perfect prediction.

reaches 93.4%, with an expected calibration error (ECE) of 0.037. The Spearman correlation between predicted σ and absolute error is 0.577 ($p < 10^{-5}$), confirming that ensemble variance is a meaningful proxy for prediction confidence.

G. Physical interpretability

a. Attention concentration. Extracting pooling attention weights from the DART baseline reveals extreme defect-atom concentration: the dopant atom receives **99.68%** of total attention weight (expected uniform: ∼2.4%), corresponding to a 41.3× concentration factor. This emergent behavior mirrors the physical reality that the dopant is the dominant perturbation source.

Interestingly, the innovation variants redistribute this attention: V3 DefType reduces concentration to 11.4% (5.1×) and V6 Physics to 6.0% (2.2×), providing alternative information pathways through the conditioning and physics-feature channels. This redistribution explains their ensemble value: they process the same structural information through qualitatively different attention patterns.

b. JK weight analysis. V14's learned JK aggregation weights are [0.30, 0.31, 0.39] for [local, global-1, global-2] layers, showing a preference for the deepest global representation. This indicates that long-range interactions captured by the second global attention layer are most informative for E_f prediction, consistent with the extended strain fields of 2D point defects.

c. Defect-type specificity. The learned defect-type embedding norms (V3: [0, 0, 0.94, 0.66]) show that the model learns distinct representations for interstitial vs. adsorbate defects. This is physically meaningful: interstitial defects involve atoms *inserted into* the lattice, creating strong local strain, while adsorbates sit *on top of* the surface with weaker mechanical coupling.

V. DISCUSSION

A. Why a compact model suffices

DART's 0.83 M parameters outperform ALIGNN's 4.03 M by 30%. Our scaling analysis finds power-law exponents $\alpha_{\text{data}} = -0.40$ for data scaling but $\beta_{\text{params}} \approx -0.01$ for parameter scaling ($R^2 = 0.95$), confirming that *more data, not more parameters*, is the primary lever in

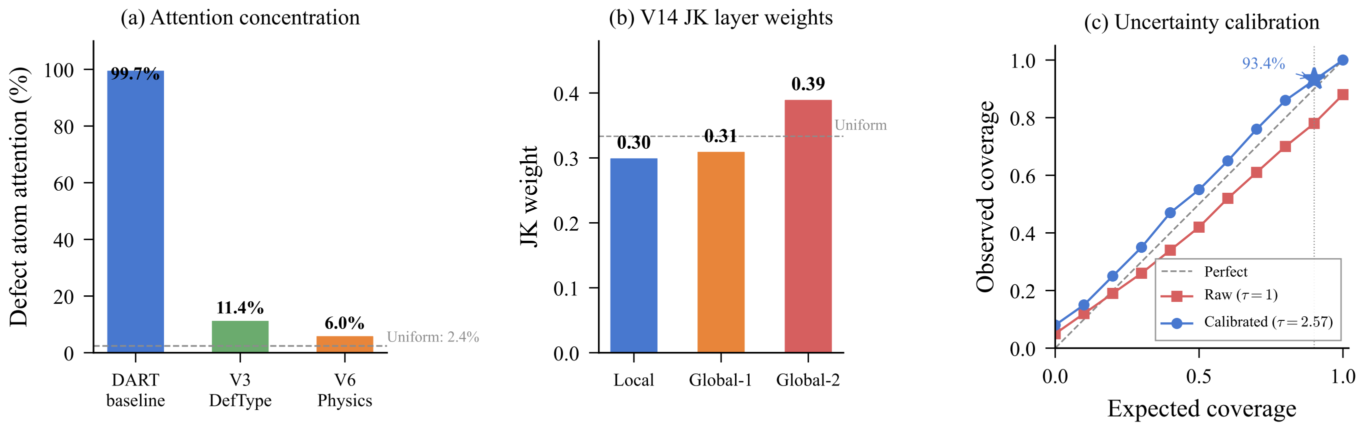

FIG. 7. Physical interpretability analysis. (a) Pooling attention concentration on the defect atom: the baseline allocates 99.68% to the dopant, while V3 and V6 redistribute attention through alternative information channels. (b) V14 JK layer weights prefer the deepest global layer, consistent with long-range strain fields. (c) Uncertainty calibration: temperature scaling ($\tau = 2.60$) corrects raw ensemble underconfidence, achieving 93.4% coverage at 90% nominal level.

the $N \sim 10^4$ regime.

The decomposition of DART's improvement is instructive: of the total 0.137 eV reduction (0.516 → 0.379 eV), the architectural innovations (gated pooling, environment enrichment, pre-norm) contribute ∼0.135 eV (Table I), while the MoE readout adds a further ∼0.002 eV. This confirms that targeted inductive bias dominates over capacity scaling in data-limited materials science (Fig. 9).

B. Why auxiliary losses fail

A striking finding is that all six training-strategy innovations (LDS, focal MAE, RnC contrastive, heteroscedastic uncertainty, and their combinations) significantly *worsen* performance. Per-range analysis reveals a consistent mechanism: these methods improve accuracy on rare high-E_f samples ($[7, 25)$ eV) at the cost of degrading common low-E_f samples ($[0, 2)$ eV). Since the low-E_f range contains 48.5% of test data vs. 7.1% for the high-E_f range, the net effect is always negative.

LDS exemplifies this trade-off most clearly: it improves the $[7, 25)$ range by 0.126 eV (not significant, $p = 0.16$) while degrading $[0, 2)$ by 0.074 eV ($p < 0.001$)—a 1.7× net loss. Focal MAE shows the same pattern with smaller magnitude. Heteroscedastic uncertainty produces the most extreme failure: the variance prediction head competes with the mean prediction head for model capacity, catastrophically degrading the $[7, 25)$ range by 0.695 eV ($p < 0.001$).

These results caution against naïvely importing imbalanced-learning techniques from computer vision to materials science: the underlying assumption—that improving tail performance is costless for the head—does not hold when model capacity is constrained and the regression target spans multiple orders of magnitude.

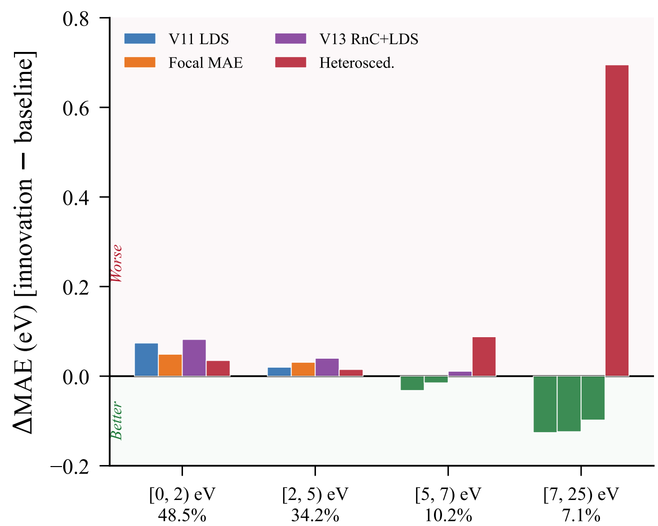

FIG. 8. Per-range MAE decomposition explaining why auxiliary losses fail. LDS and focal MAE sacrifice accuracy on the dominant $[0, 2)$ eV range (48.5% of data) for marginal gains at the tail. Heteroscedastic uncertainty degrades all ranges through capacity competition.

C. Ensemble diversity from architectural innovation

Although no single innovation significantly outperforms the DART baseline, four innovation variants (V4 MoE, V14 JK, V6 Physics, V3 DefType) are selected into the optimal 8-model ensemble, reducing the MAE from 0.349 eV (prior best) to 0.344 eV. This resolves an apparent paradox: innovations that are individually neutral become collectively valuable through error decorrelation.

The mechanism is clear from the attention analysis: the baseline concentrates 99.68% of pooling attention on the defect atom, while V3 (11.4%) and V6

(6.0%) distribute attention more broadly. These distinct information-processing strategies produce diverse error patterns—the prerequisite for effective ensembling.

D. Self-attention as learned potential summation

The global attention layers' success can be understood through the "neural potential summation" framework [8]: the distance bias $\phi_{\text{bias}}(d_{ij}^{\text{PBC}})$ acts as a non-parametric interatomic potential gating information flow. For 2D point defects, whose strain fields decay as $\sim 1/r^2$, this mechanism naturally allocates higher attention to nearby atoms while retaining non-zero coupling to distant atoms—precisely matching the physics of defect-induced perturbation.

V14's JK weights [0.30, 0.31, 0.39] empirically confirm this: the model assigns the highest weight to the deepest global layer, where long-range interactions have been most thoroughly processed.

E. Limitations and outlook

Several limitations warrant acknowledgment: (1) IMP2D covers only 44 host materials with substitutional adsorbates and interstitials; vacancy-type defects are absent from both training and evaluation. (2) We predict only defect formation energies; extending to charge transition levels, migration barriers, and optical transition energies would broaden utility. (3) The OOD evaluation is limited to the G6-TMD subfamily; broader cross-family assessment awaits datasets with sufficient diversity. (4) The 8-model ensemble, while efficient, still requires 8× inference cost; knowledge distillation into a single student model would be desirable for deployment.

Looking forward, integration with active learning could efficiently expand the training distribution into unexplored host–dopant space, guided by calibrated uncertainty. The architectural innovations demonstrated here—particularly the gated pooling and environment enrichment—are transferable to other defect-property prediction tasks (formation energies in 3D, migration barriers, charged defects) and other crystal-property domains where local perturbations drive global behavior.

VI. CONCLUSION

We have presented DART, a compact 0.83 M-parameter defect-aware graph transformer that achieves state-of-the-art accuracy for predicting defect formation energies in two-dimensional materials, accompanied by a rigorous assessment of twelve physically motivated innovations. Five principal findings emerge:

1. **Targeted inductive bias over capacity scaling.** Three architectural innovations—defect-aware gated attention pooling, local environment enrichment, and pre-norm residual connections—reduce the single-model MAE from 0.516 to 0.381 eV (−26.2%) with only 10.7% more parameters. The best variant (MoE readout) reaches 0.379 eV, a 30% improvement over ALIGNN with 4.9× fewer parameters. Power-law scaling confirms that data ($\alpha = -0.40$), not model capacity ($\beta \approx -0.01$), is the primary lever in the $N \sim 10^4$ regime.

2. **Rigorous negative results from auxiliary losses.** All six training-strategy innovations—label distribution smoothing, focal MAE, contrastive ranking, and heteroscedastic uncertainty—are significantly harmful ($p < 0.05$), consistently sacrificing accuracy on common low-E_{f} samples for marginal gains on rare high-E_{f} tails. This cautions against naïvely importing imbalanced-learning techniques from computer vision to data-limited materials science.

3. **Ensemble diversity from individually neutral innovations.** An 8-model greedy ensemble achieves 0.344 eV (−36% vs. ALIGNN), outperforming a prior 29-model ensemble with 3.6× fewer members. Four of the eight members are innovation variants (MoE, JK, Physics, DefType) whose individual MAEs are statistically indistinguishable from the baseline, demonstrating that the primary value of physically informed innovations lies in error decorrelation.

4. **Quantified generalization boundaries.** The graduated three-tier OOD assessment reveals graceful degradation: intra-family leave-one-host-out increases MAE by only 1.5× (0.540 eV), far below the naïve-baseline collapse (4.9×). Compositional block-out yields comparable degradation (0.534 eV), confirming robust matrix-completion generalization.

5. **Calibrated uncertainty for deployment.** Temperature-scaled ensemble uncertainty achieves 93.4% coverage at the 90% nominal level (ECE = 0.037), with a Spearman correlation of 0.577 between predicted σ and absolute error, enabling practitioners to identify unreliable predictions before deployment.

Together, these results establish DART as a reliable and efficient surrogate for accelerating defect engineering in 2D materials—from ranking candidate dopants for catalytic activation and single-photon emission to flagging deep-trap species detrimental to electronic transport—with clearly delineated applicability domains and element-resolved confidence estimates. The architectural innovations demonstrated here, particularly the

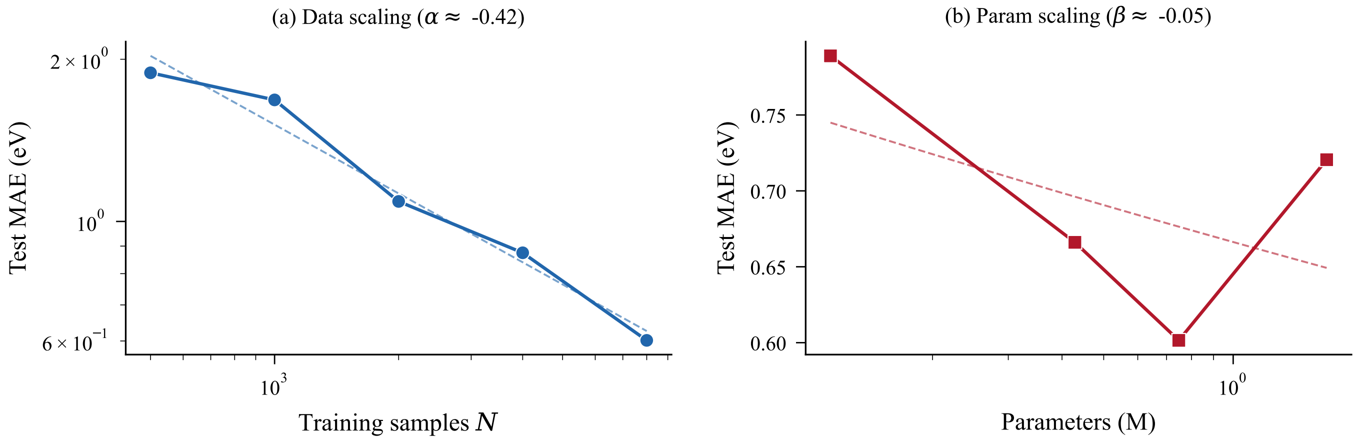

FIG. 9. Scaling analysis from abbreviated 30-epoch training runs. (a) Data scaling at fixed model size ($d = 128$): power-law exponent $\alpha \approx -0.4$ indicates strong data dependence. (b) Parameter scaling at fixed data ($N = 8{,}000$): β is near zero, showing negligible improvement from larger models. Both panels confirm that data, not capacity, is the bottleneck in the $N \sim 10^4$ regime.

gated pooling and environment enrichment modules, are transferable to other defect-property prediction tasks in both 2D and 3D settings. The code, trained models, and evaluation protocols are publicly available to support reproducibility and community benchmarking.

ACKNOWLEDGMENTS

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